

SMART FACTORY ARCHITECTURE

From Technology to Implementation —

In this session, we will explore the architecture of a modern Smart Factory and see how Industry 4.0 technologies are integrated to deliver real-time visibility, intelligence and business value.

How do all these technologies come together inside a real Industry 4.0 environment?



SMART FACTORY – THE HEART OF INDUSTRY 4.0

A highly intelligent ecosystem where physical assets, advanced sensors, software systems, human operators, and core business processes are completely connected.



TRADITIONAL FACTORY

Isolated. Delayed. Inefficient.



VS

SMART FACTORY

Connected. Real-time. Intelligent.



KEY BENEFITS

-  Unprecedented Operational Efficiency
-  Elevated Product Quality
-  Maximum Manufacturing Agility
-  Smarter, Data-Backed Decisions

FROM DATA TO ACTIONABLE INTELLIGENCE



 Driving Excellence Across People, Processes & Performance

SMART FACTORY ARCHITECTURE

From Technology to Implementation

In this session, we will explore the architecture of a modern Smart Factory and see how Industry 4.0 technologies are integrated to deliver real-time visibility, intelligence and business value.



AGENDA

- 1 Key Building Blocks of a Smart Factory
- 2 Shop Floor Layer: Devices, Sensors & Equipment
- 3 Connectivity & Communication Networks
- 4 Edge & Cloud Integration
- 5 Data Management, Analytics & AI
- 6 Enterprise Systems Integration (MES, ERP, SCM)
- 7 Real-time Monitoring & Decision Making
- 8 End-to-End Data Flow: From Machines to Business Outcomes
- 9 Key Benefits of Smart Factory Architecture
- 10 Summary & Key Takeaways



SETTING THE CONTEXT

The Smart Factory connects **people, machines, data** and **processes** through integrated technologies and intelligent systems.



Machines
generate
data



Networks
move
data



Platforms
process
data



AI
creates
intelligence



**Business
Systems**
convert intelligence
into action



A Smart Factory is a living digital ecosystem that delivers agility, efficiency, quality, and competitive advantage.



LEARNING OUTCOMES

- ✓ Understand the layers and building blocks of a Smart Factory
- ✓ Learn how data flows from the shop floor to enterprise systems
- ✓ See how technologies work together to create intelligence and drive actions
- ✓ Understand the role of real-time data, analytics and AI in decision making
- ✓ Recognize the business impact of a well-architected Smart Factory



OUR GOAL:

By the end of this session, you will be able to understand how a Smart Factory is architected, how data flows across the system, and how organizations turn data into real business value.



**BUILDING SMARTER FACTORIES.
CREATING A BETTER FUTURE.**

SMART FACTORY ARCHITECTURE: LAYERS & BUILDING BLOCKS

A multi-layered architecture that connects machines to people and transforms data into intelligent actions.



ARCHITECTURE GOAL: Connect everything, collect the right data, generate intelligence and enable real-time actions to create a Smart, Resilient and Sustainable Factory

SHOP FLOOR / PHYSICAL LAYER

The Foundation of the Smart Factory

The shop floor is where real-world operations happen and data is born.



Real Assets
Machines, tools
and equipment



Real Data
Captured from
the physical world



Real Operations
Manufacturing
in action



Real Impact
Quality, efficiency
and productivity

OVERVIEW

The Shop Floor / Physical Layer includes all physical assets, devices and systems that perform operations and generate raw data. This layer is the starting point of the data journey in a Smart Factory.

KEY FUNCTIONS



Perform real-world manufacturing and operational tasks



Capture accurate, real-time data from assets and environment



Ensure quality, safety and compliance on the shop floor



Provide visibility into machine status and process conditions



Enable automation and closed-loop control at the source

KEY COMPONENTS

SENSORS & TRANSDUCERS



Measure physical parameters such as temperature, pressure, vibration, speed, position, flow, level, and more.

ACTUATORS & DRIVES



Convert control signals into mechanical action to start, stop, move or adjust operations.

MACHINES & EQUIPMENT



Production assets that perform cutting, assembling, welding, packing, inspection and other tasks.

PLCS & CONTROLLERS



Execute control logic, collect data from devices and ensure precise, reliable operations.

ROBOTS & COBOTS



Automate repetitive tasks, improve precision, flexibility and safety.

ENVIRONMENT & UTILITY SYSTEMS



Monitor and manage environmental conditions and utilities for optimal operations.



This layer captures the pulse of the factory and sends it to upper layers for processing, analysis and action.

DATA GENERATED

- Machine status (On/Off, Idle, Fault)
- Production counts and cycle times
- Process parameters (Temp, Pressure, Speed)
- Quality data (Measurements, Defects)
- Energy consumption
- Alarms and events
- Environmental conditions

EXAMPLES ON THE SHOP FLOOR



A temperature sensor on an oven monitors heat in real time.



A robot arm picks and places parts on the assembly line.



A PLC controls the sequence of machines in a production cell.



A vision system inspects products for defects.



A drive controls the speed of a conveyor belt.



KEY TAKEAWAY

The Shop Floor / Physical Layer is the foundation of Industry 4.0. It generates the raw data that fuels the entire Smart Factory ecosystem.



Real World
Operations



Data
Captured



Data Sent
Upstream



Intelligence
Generated



Value
Delivered

CONNECTIVITY LAYER: SECURELY CONNECT & TRANSMIT DATA

Building the Digital Highways of the Smart Factory

The Connectivity Layer ensures reliable, secure and real-time communication between devices, systems and applications.



OVERVIEW



This layer connects devices on the shop floor to edge systems, cloud platforms and enterprise applications. It enables fast, reliable and secure data exchange across the entire factory.

KEY FUNCTIONS



Enable reliable and real-time communication



Ensure secure data transmission



Connect OT systems to IT and cloud



Support a variety of industrial protocols and networks



Provide scalability, redundancy and high availability

CONNECTIVITY COMPONENTS

INDUSTRIAL NETWORKS



Ethernet, PROFINET, EtherNet/IP, Modbus TCP and other industrial networks connect machines, PLCs and devices within the factory.

IOT PROTOCOLS



MQTT, OPC UA, AMQP, CoAP and HTTP/HTTPS enable efficient and standardized data exchange between devices and platforms.

GATEWAYS & ROUTERS



Gateways collect data from local devices, perform protocol translation and route data to edge or cloud systems.

WIRELESS CONNECTIVITY



Wi-Fi, Bluetooth, Zigbee, LoRaWAN and Cellular (4G/5G) provide flexible connectivity across machines and facilities.

NETWORK SECURITY



Firewalls, VPN, IDS/IPS and secure access controls protect data in transit and prevent unauthorized access across the network.

TIME & DATA SYNCHRONIZATION



NTP and time sync mechanisms ensure accurate timestamps for data consistency, correlation and analytics.

COMMON COMMUNICATION OPTIONS



Wired: Ethernet, Industrial Ethernet, Fiber Optics



Wireless: Wi-Fi, 5G, LoRaWAN, Bluetooth



Fieldbus: PROFIBUS, CANopen, DeviceNet



LPWAN: LoRaWAN, NB-IoT for long range communication

KEY OUTCOMES

- ✓ Seamless data flow across OT and IT
- ✓ Low latency and high reliability
- ✓ Secure and trusted communications
- ✓ Supports real-time monitoring and control
- ✓ Enables scalability and future growth

KEY TAKEAWAY

The Connectivity Layer acts as the digital highway of the Smart Factory, ensuring that the right data reaches the right place at the right time.



Data from Devices



Transmit Securely



To Edge / Cloud



For Analytics & Applications



For Better Decisions

EDGE LAYER: PROCESS DATA CLOSER TO THE SOURCE

Enabling Real-time Insights and Faster Decisions

The Edge Layer processes data near the source to reduce latency, save bandwidth and enable real-time actions.



OVERVIEW



The Edge Layer sits between the shop floor devices and the cloud. It performs local processing, filtering and analysis to deliver faster responses and reduce dependency on the cloud.

KEY FUNCTIONS



Process data in real time near the source



Filter and aggregate data to reduce volume



Trigger instant alerts and automated actions



Ensure local reliability even with limited connectivity



Reduce bandwidth usage and cloud costs

EDGE COMPONENTS

EDGE DEVICES & CONTROLLERS



Industrial PCs, rugged devices and micro-servers that collect and process data locally.

LOCAL DATA PROCESSING



Execute analytics, apply rules and run applications at the edge for quick decision-making.

DATA FILTERING & AGGREGATION



Remove unnecessary data, aggregate information and send only valuable data upstream.

EDGE ANALYTICS & AI



Run AI/ML models locally for anomaly detection, prediction and optimization.

REAL-TIME ACTIONS & CONTROL



Trigger control actions, alerts and workflows based on local insights.

STORE & FORWARD BUFFERING



Store data temporarily when connectivity is interrupted and forward later.

SECURITY AT THE EDGE



Device security, encryption, access control and secure communication.

EDGE BENEFITS



Ultra-low latency and faster response



Reduced bandwidth and costs



Reliable operations even with poor connectivity



Improved data privacy and security



Enables real-time control and automation



Better performance for critical applications

EDGE IN ACTION: DATA FLOW

SHOP FLOOR DEVICES



Machines, sensors and PLCs generate data.

EDGE GATEWAY / EDGE DEVICE



Data is collected and processed at the edge.

EDGE PROCESSING



Filter, analyze and generate insights locally.

ACTIONS & INSIGHTS



Real-time actions, alerts and control at the source.

CLOUD / DATA CENTER (UPSTREAM)



Send only valuable, summary data to cloud.



KEY TAKEAWAY

The Edge Layer brings intelligence closer to the source, enabling real-time decisions, reducing latency and ensuring resilient operations in a Smart Factory.



Data Generated at Source



Processed at Edge



Actions in Real Time



Insights to Cloud



Business Value

USE CASE EXAMPLES



Predictive maintenance on shop floor machines



Real-time quality inspection and defect detection

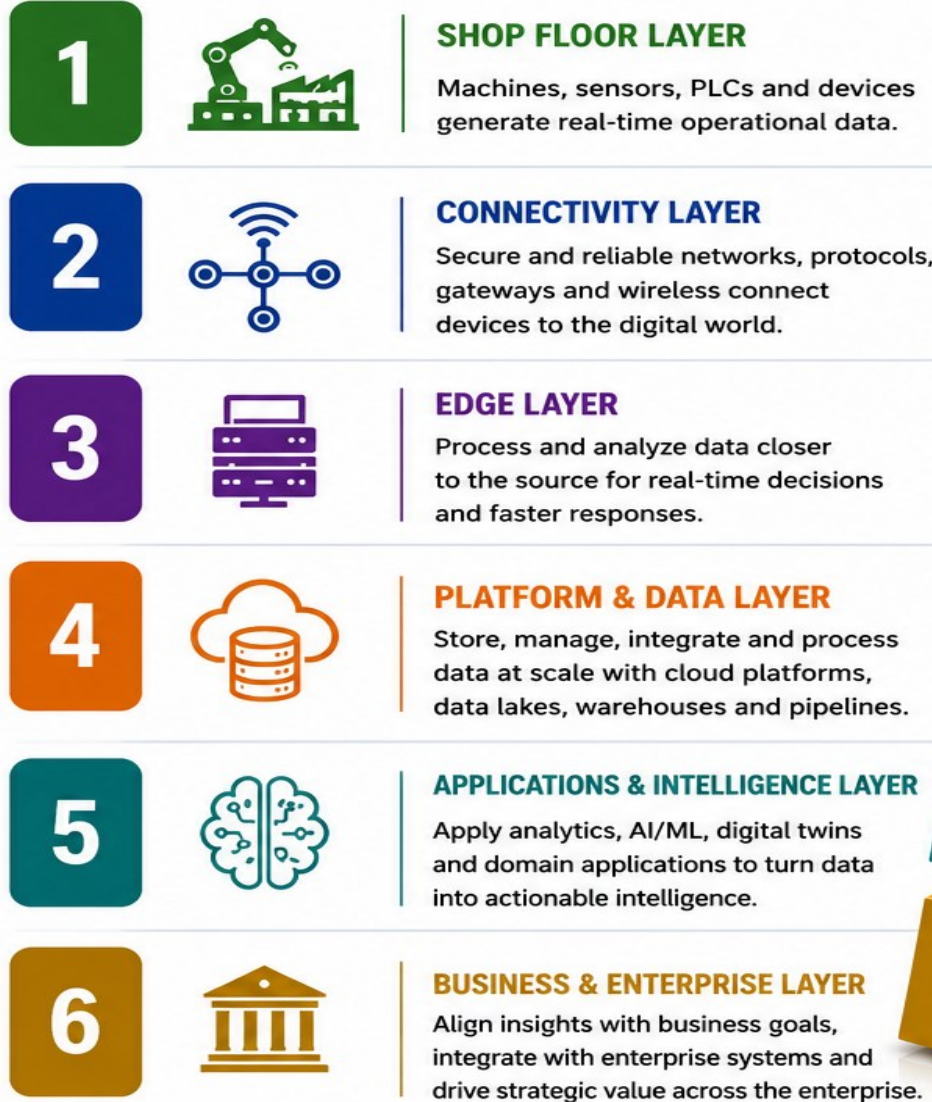


Instant safety alerts and equipment shutdown



Energy optimization and load management

THE 6 LAYERS OF SMART FACTORY ARCHITECTURE



END-TO-END DATA & VALUE FLOW



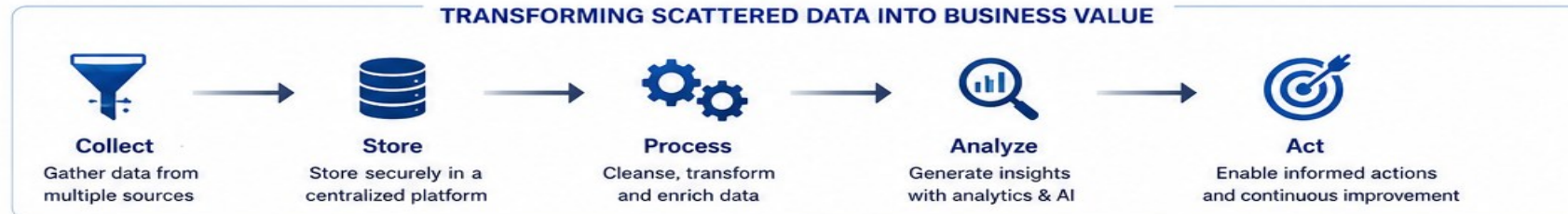
PLATFORM & DATA LAYER

The Digital Foundation of the Smart Factory

DATA COMES FROM ACROSS THE ENTERPRISE



TRANSFORMING SCATTERED DATA INTO BUSINESS VALUE



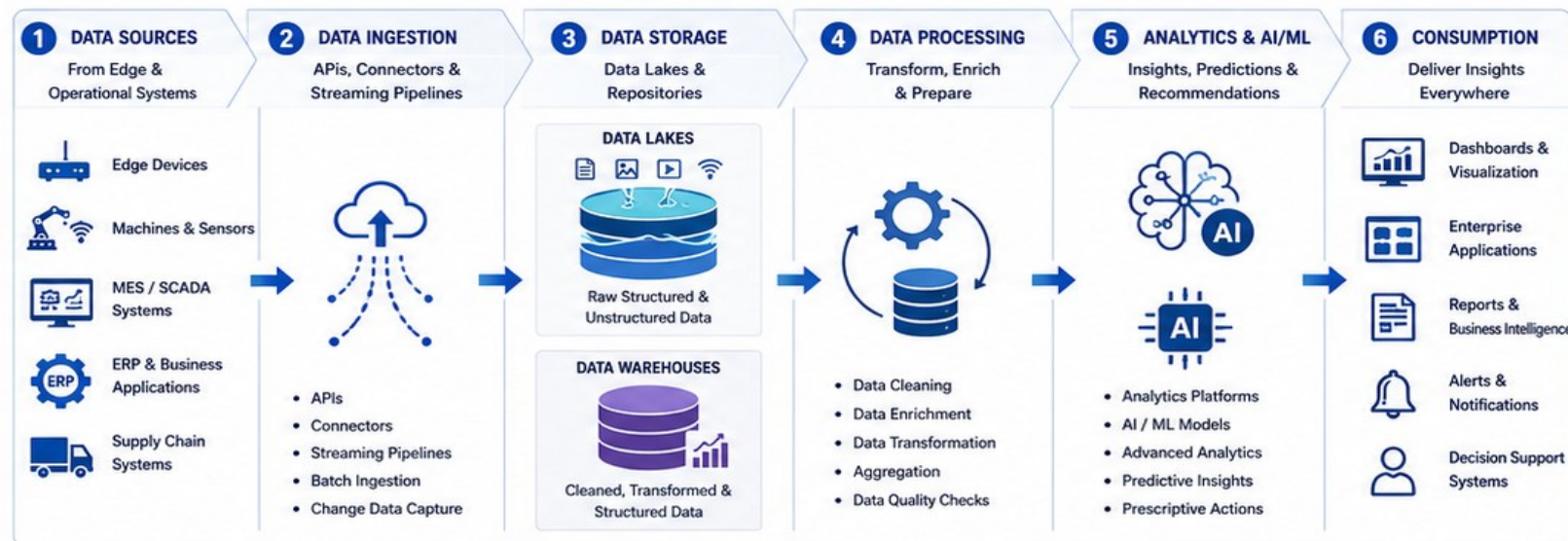
PLATFORM & DATA LAYER – CAPABILITIES, DATA FLOW & BUSINESS VALUE

Transforming Raw Operational Data into a Strategic Business Asset

KEY CAPABILITIES

- Cloud & Data Platforms**
Scalable infrastructure for storing and processing large volumes of information.
AWS, Microsoft Azure, Google Cloud, Private Cloud
- Data Lakes**
Repository for raw structured and unstructured data. Store information at virtually any scale.
- Data Warehouses**
Store cleaned and transformed data optimized for reporting, dashboards, and business intelligence.
- Stream Processing**
Process high-velocity data in real time and generate immediate insights.
- Data Integration & ETL**
Bring together data from multiple sources (Machines, Sensors, ERP, MES, SCM, etc.) for unified analysis.
- Data Catalog, Metadata & Governance**
Classify, secure, and manage data throughout its lifecycle to ensure trust, consistency, and compliance.

END-TO-END DATA FLOW



FOUNDATION ENABLERS (Across All Stages)

- Security & Access Management
- Data Governance & Policies
- Metadata Management
- Data Quality Management
- Monitoring & Observability
- Compliance & Audit

COMMON USE CASES

- Production Analytics
- Predictive Maintenance
- Quality Management
- Supply Chain Optimization
- Energy Management
- Executive Dashboards

BUSINESS VALUE

- Centralized Data Management**
Single source of truth across the enterprise.
- Better Data Quality**
Accurate, consistent, and trusted data.
- Stronger Security & Compliance**
Protect data and meet regulatory requirements.
- Faster Analytics & Insights**
Real-time and historical insights to drive performance.
- Data-Driven Decision Making**
Empower teams to make smarter faster decisions.
- Scalability & Agility**
Handle growing data volumes and evolving business needs.



KEY TAKEAWAY

The Platform and Data Layer transforms raw operational data into a strategic business asset.



Without This Layer:

Data remains fragmented across systems.



With This Layer:

Organizations gain a single source of truth that powers analytics, AI, and intelligent decision-making throughout the Smart Factory.

The Applications and Intelligence Layer.

The real value emerges when data is transformed into intelligence, insights, and actions.

That is exactly the role of the Applications and Intelligence Layer.

Think of this layer as the brain of the Smart Factory. It analyzes data, identifies patterns, predicts future events, recommends actions, and enables smarter decision-making across the organization.

LEVELS OF INTELLIGENCE

1 DESCRIPTIVE ANALYTICS



What happened?

Using reports, dashboards, and KPIs, organizations gain visibility into production performance, equipment utilization, quality metrics, and operational outcomes.

2 DIAGNOSTIC ANALYTICS



Why did it happen?

By analyzing trends, correlations, and root causes, organizations can identify the factors contributing to a problem or performance issue.

3 PREDICTIVE ANALYTICS



What is likely to happen next?

Using statistical models and Machine Learning algorithms, organizations can forecast equipment failures, production bottlenecks, quality issues, and demand fluctuations.

4 PRESCRIPTIVE ANALYTICS



What should we do about it?

The system not only predicts future events but also recommends the best course of action to optimize outcomes.

INTELLIGENCE ENABLERS (THE CORE ENGINE)



MACHINE LEARNING & ARTIFICIAL INTELLIGENCE MODELS

Continuously learn from historical and real-time data, becoming smarter over time and improving decision quality.



DIGITAL TWINS & SIMULATION MODELS

Virtual replicas allow organizations to test scenarios, optimize operations, and evaluate decisions before implementing them.



VISUALIZATION & DELIVERY PLATFORM

Insights delivered through dashboards, visualization platforms, mobile apps, and operational systems—making intelligence accessible to everyone across the enterprise.

BUSINESS VALUE



Higher productivity



Improved product quality



Reduced downtime



Lower operational costs



Better resource utilization



Greater sustainability



Faster, data-driven decision-making

FROM DATA TO INTELLIGENCE TO BUSINESS VALUE



1. DATA GENERATION

Machines, sensors and systems generate operational data.



2. PLATFORM LAYER

Data is collected, stored, integrated and processed at scale.



3. INTELLIGENCE LAYER

Data is analyzed, patterns are discovered, predictions and recommendations are generated.



4. ACTION & AUTOMATION

Recommendations trigger actions, workflows or operational adjustments.

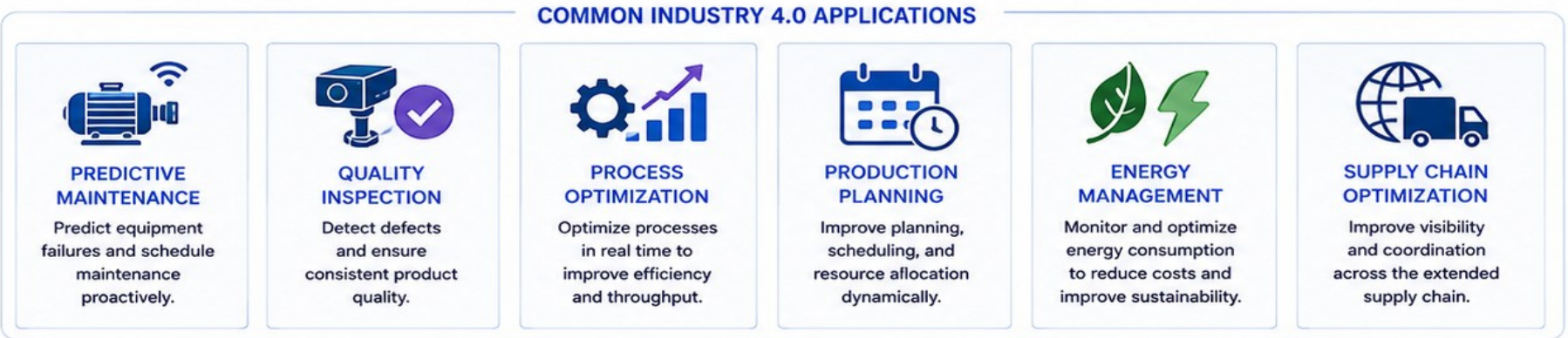
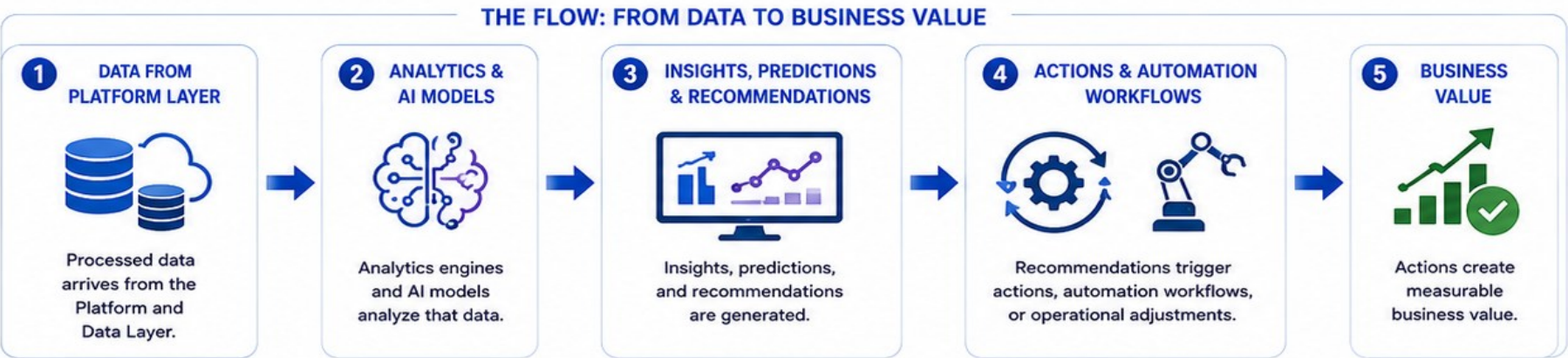


5. BUSINESS OUTCOMES

Actions create measurable impact and drive sustainable business value.

The Applications and Intelligence Layer in Action

Transforming Data into Intelligence, Actions, and Business Value



KEY TAKEAWAY

Data becomes intelligence at this layer.

This is where organizations move beyond monitoring and reporting to prediction, optimization, and autonomous decision-making.



Business & Enterprise Layer: Overview

Connecting Intelligence to Business Value

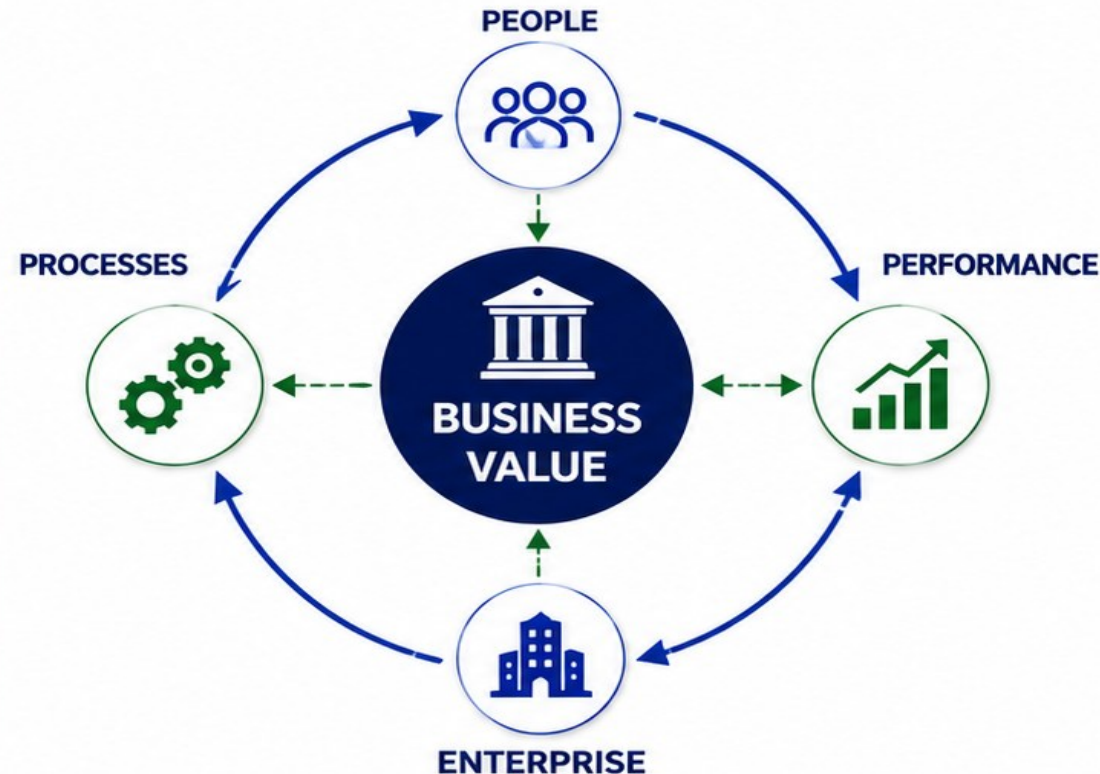
The **Business & Enterprise Layer** connects insights from the Intelligence Layer with enterprise systems, business processes, and people to create measurable business value.

OVERVIEW



This layer ensures that operational intelligence is aligned with business objectives, governance requirements, financial goals, and strategic priorities.

It turns operational insights into strategic business outcomes.



KEY FUNCTIONS



Align operations with business strategy



Integrate enterprise applications and data



Enable collaboration across departments and functions



Monitor performance through KPIs and dashboards



Support governance, compliance, and risk management



Drive continuous improvement and value realization



Turning Operational Intelligence into Strategic Business Outcomes



Business & Enterprise Layer: Key Enterprise Capabilities

Transforming Insights into Enterprise-Wide Decisions and Actions

The **Business & Enterprise Layer** enables organizations to integrate systems, orchestrate processes, manage performance, ensure compliance and drive continuous value across the enterprise.

1



ENTERPRISE SYSTEM INTEGRATION

Connect ERP, MES, CRM, SCM, PLM and other enterprise systems for seamless information flow.

2



BUSINESS INTELLIGENCE & REPORTING

Provide dashboards, reports, scorecards and insights for executive visibility and better decision-making.

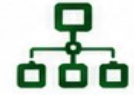
3



STRATEGIC PLANNING & FORECASTING

Use analytics and AI to forecast demand, plan capacity, optimize resources and achieve financial and operational goals.

4



WORKFLOW & PROCESS ORCHESTRATION

Automate and orchestrate end-to-end business processes across functions, departments and locations.

5

PERFORMANCE MANAGEMENT

Monitor KPIs and OKRs across operations, finance, quality, safety and other key areas to drive performance.

6

RISK, COMPLIANCE & GOVERNANCE

Ensure regulatory compliance, data governance, auditability, security and enterprise-wide risk management.

7

VALUE REALIZATION & CONTINUOUS IMPROVEMENT

Measure outcomes, identify improvement opportunities and drive innovation to maximize enterprise value.



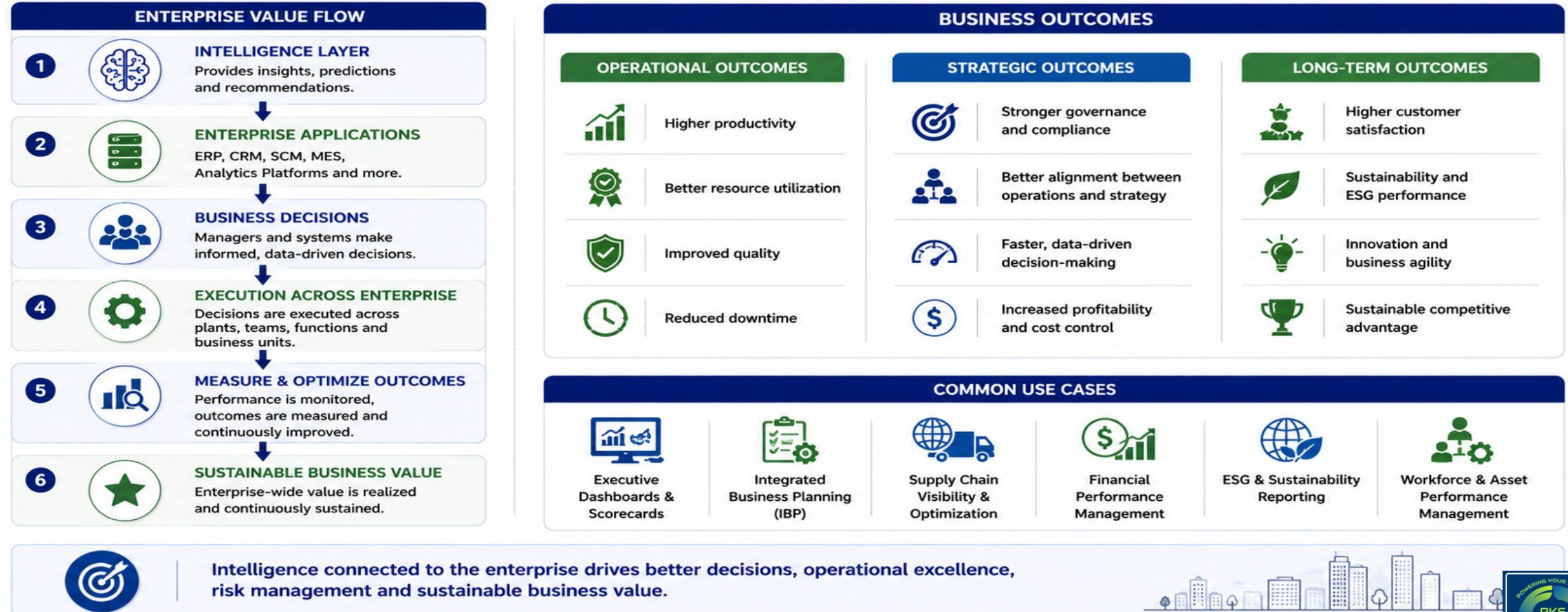
These capabilities ensure that intelligence is aligned with business objectives and translated into actions that deliver measurable and sustainable business value.



Business & Enterprise Layer: Enterprise Value Flow & Outcomes

How Intelligence Creates Business Impact

The Business & Enterprise Layer converts insights into decisions and actions, executes them across the enterprise, and continuously measures outcomes to create sustainable business value.



BUSINESS & ENTERPRISE LAYER: ALIGN OPERATIONS WITH BUSINESS GOALS

Integrate, Orchestrate and Deliver Value Across the Enterprise

This layer connects insights and applications to enterprise systems and people to drive collaboration, governance and strategic decision-making.



OVERVIEW

The Business & Enterprise Layer ensures that intelligence from across the Smart Factory is aligned with business objectives, processes and strategy to create sustainable value and competitive advantage.

KEY FUNCTIONS

- Align operations with business strategy
- Integrate IT and OT systems and data
- Orchestrate processes across the enterprise
- Enable collaboration across business functions
- Track performance against KPIs and goals
- Ensure governance, risk management and compliance
- Support financial planning and value realization

KEY ENTERPRISE CAPABILITIES

ENTERPRISE SYSTEM INTEGRATION



Seamless integration with ERP, MES, PLM, SCM, CRM, CMMS and other enterprise systems.

BUSINESS INTELLIGENCE & REPORTING



Consolidate data and intelligence to provide executive dashboards, reports and scorecards in real time.

STRATEGIC PLANNING & FORECASTING



Use analytics and AI to forecast demand, capacity, resources and financial outcomes.

WORKFLOW & PROCESS ORCHESTRATION



Automate and orchestrate end-to-end business processes across departments and locations.

PERFORMANCE MANAGEMENT



Monitor KPIs, OKRs and performance across operations, finance, quality, safety and more.

RISK, COMPLIANCE & GOVERNANCE



Ensure regulatory compliance, data governance, auditability and enterprise-wide risk management.

VALUE REALIZATION & CONTINUOUS IMPROVEMENT



Drive continuous improvement, innovation and measurable business outcomes.

ENTERPRISE VALUE FLOW

INSIGHTS FROM INTELLIGENCE LAYER



Actionable insights, predictions and recommendations.

ENTERPRISE APPLICATIONS



Insights are consumed by enterprise systems and applications.

BUSINESS PROCESSES & DECISIONS



People and systems collaborate to make informed decisions.

EXECUTION ACROSS THE ENTERPRISE



Decisions are executed across plants, functions and geographies.

MEASURE & OPTIMIZE OUTCOMES



Outcomes are measured and optimized for maximum business impact.

SUSTAINABLE BUSINESS VALUE



Improved resilience, agility, profitability and customer satisfaction.

TYPICAL ENTERPRISE SYSTEMS



BUSINESS OUTCOMES

- Enterprise-wide visibility and transparency
- Better alignment between operations and strategy
- Higher efficiency and operational excellence
- Improved profitability and cost control
- Stronger compliance and risk management
- Higher customer satisfaction and experience
- Sustainable growth and long-term value

COMMON USE CASES

- Executive dashboards & scorecards
- Integrated business planning (IBP)
- Supply chain visibility & optimization
- Financial performance management
- Sustainability & ESG reporting
- Workforce and asset performance

KEY TAKEAWAY

The Business & Enterprise Layer ensures that intelligence is connected to people, processes and systems—turning insights into actions that drive strategic business value.



Intelligence Generated



Aligned with Business Goals



Integrated & Orchestrated



Governed & Managed

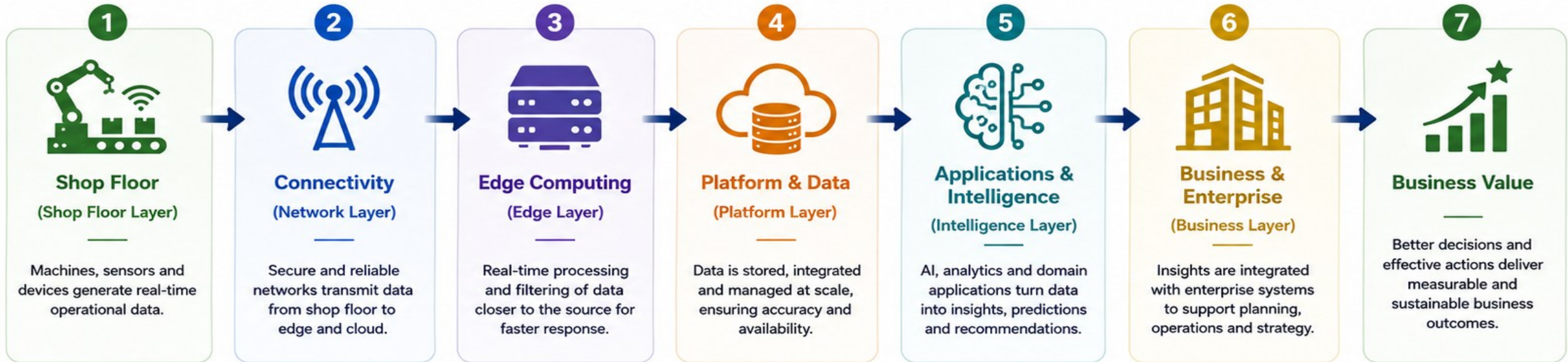


Business Value Realized

Smart Factory Architecture: End-to-End Value Flow

From Machines to Meaningful Business Outcomes

Data flows across the Smart Factory architecture, is transformed into intelligence, and drives decisions and actions that create measurable business value.



KEY BENEFITS



Connected Data + Intelligent Systems + Business Alignment = Sustainable Business Value